The rank-1-leak decomposition: a structural proof of σ_2-G1 for the rank-2 shapes at $n = 6$

Clio

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Abstract

We close the open structural step in the σ_2-G1 program at $n = 6$. For each of the two non-trivial rank-2 shapes $\lambda \in \{(5,1),(3,2,1)\}$, we exhibit an explicit decomposition

$$\sigma_2\big(\Pi_q^{(6)} \mid V_\lambda\big) = \sum_i \frac{q^{a_i}\,(1+q)^{b_i}\,P_i(q)}{\Phi_4(q)},$$

with $a_i, b_i \in \mathbb{Z}_{\geq 0}$ and each $P_i(q)$ a palindromic polynomial with non-negative integer coefficients. Each summand is manifestly non-negative for $q > 0$, except for the common cyclotomic denominator $\Phi_4(q) = q^2 + 1$, whose cancellation is reduced in each case to a single polynomial identity which we verify directly. The resulting expression for σ_2 is therefore a manifestly palindromic polynomial with non-negative integer coefficients, settling σ_2-G1 for the two non-trivial rank-2 shapes at $n = 6$.

The third rank-≤ 2 shape $\lambda = (2, 1^4)$ is rank-0 (its H_6-invariants vanish), so $\sigma_2 = 0$ trivially. All remaining $\lambda \vdash 6$ are not rank-2, hence outside the scope of this note.

1 Setup

Let $H_q = H_q(\mathfrak{S}_6)$ be the Hecke algebra over $\mathbb{Z}[q]$ with generators $T_1, \ldots, T_5$ subject to the quadratic relation $T_i^2 = (q-1)T_i + q$ and the standard braid/commutation relations. Let

$$\underline{w}_0 = (1;\ 2,1;\ 3,2,1;\ 4,3,2,1;\ 5,4,3,2,1)$$

be the staircase reduced word for the longest element $w_0 \in \mathfrak{S}_6$ (length $\binom{6}{2} = 15$), and define the *staircase Bott–Samelson element*

$$\Pi_q^{(6)} \coloneqq \prod_{j=1}^{15} (1 + T_{i_j}) \in H_q.$$

For each partition $\lambda \vdash 6$, V_λ denotes the corresponding cell module (Specht module). We write $\sigma_k(\Pi_q^{(6)} \mid V_\lambda)$ for the k-th elementary symmetric function in the eigenvalues of $\Pi_q^{(6)}$ on V_λ, equivalently the trace of $\Lambda^k \Pi_q^{(6)}$ on $\Lambda^k V_\lambda$.

Definition 1 (σ_k-G1)**.** Write $\sigma_k(V_\lambda) = q^a (1+q)^b\, P_{\lambda,k}(q)$ with (a,b) maximal. We say σ_k-G1 holds for λ if $P_{\lambda,k}(q) \in \mathbb{Z}_{\geq 0}[q]$.

Definition 2 (Rank of V_λ)**.** Let $H_n \coloneqq \langle s_1, s_3, s_5, \ldots, s_{n-1} \rangle$ for n even (the parity-alternating parabolic). The *rank* of V_λ is $\dim V_\lambda^{H_n}$, the dimension of the simultaneous $+q$-eigenspace. By the H-invariant theorem (Apr 17), $\operatorname{rank}(\Pi_q^{(n)} \mid V_\lambda) = \dim V_\lambda^{H_n}$.

The complete rank profile for $\lambda \vdash 6$ (verified by direct computation) is

λ	(6)	$(5,1)$	$(4,2)$	$(4,1,1)$	$(3,3)$	$(3,2,1)$	$(3,1^3)$	$(2,2,2)$	$(2^2,1^2)$	$(2,1^4)$	(1^6)
$\dim V_\lambda^{H_6}$	1	**2**	3	1	1	**2**	0	1	0	0	0

Hence *the rank-2 shapes at $n = 6$ are exactly* $\{(5,1),(3,2,1)\}$; the conjugate $(2,1^4)$ has rank 0 (consistent with $\ell((2,1^4)) = 5 > (n+1)/2$), giving $\sigma_2(V_{(2,1^4)}) = 0$ trivially.

For the two remaining cases we use the Hoefsmit q-deformed seminormal form for H_q acting on V_λ; the matrices of $T_1, \ldots, T_5$ are explicit rational-function matrices over $\mathbb{Q}(q)$, and the Hecke relations are verified symbolically (`compute_filtration.py`, see Section 7).

2 Main theorem

Theorem 3 (Rank-1-leak σ_2 decomposition at $n = 6$)**.** *For each of the two non-trivial rank-2 shapes $\lambda \in \{(5,1),(3,2,1)\}$, the polynomial $\sigma_2(\Pi_q^{(6)} \mid V_\lambda) \in \mathbb{Z}[q]$ admits an explicit decomposition into Φ_4-rational pieces, each manifestly palindromic with non-negative integer numerator coefficients. Specifically:*

(a) For $\lambda = (5,1)$,

$$\sigma_2(V_{(5,1)}) = [A] + [B],$$

where

$$[A] = \frac{q^3(1+q)^{26}}{\Phi_4(q)},$$
$$[B] = \frac{q^4(1+q)^{18}\,(q^2+4q+1)\,(4q^4+9q^3+4q^2+9q+4)}{\Phi_4(q)},$$

and the cancellation

$$q^3(1+q)^{18}\big[(1+q)^8 + q(q^2+4q+1)(4q^4+9q^3+4q^2+9q+4)\big] = \Phi_4(q)\cdot\sigma_2(V_{(5,1)})$$

is a polynomial identity (Lemma 7).

(b) For $\lambda = (3,2,1)$,

$$\sigma_2(V_{(3,2,1)}) = [A_{rr}] + [A_{ee}] + [\text{cross}],$$

where

$$[A_{rr}] = \frac{q^9(1+q)^{10}\,(q^2+q+1)(q^2+4q+1)}{\Phi_4(q)},$$
$$[A_{ee}] = \frac{q^{11}(1+q)^2\,(q^2+q+1)(q^2+4q+1)(q^4+q^3+q^2+q+1)}{\Phi_4(q)},$$
$$[\text{cross}] = \frac{q^{10}(1+q)^6\,(3q^6+19q^5+36q^4+23q^3+36q^2+19q+3)}{\Phi_4(q)},$$

and the cancellation

$$\begin{aligned} q^9(1+q)^2\big[&(1+q)^8(q^2+q+1)(q^2+4q+1) + q^2(q^2+q+1)(q^2+4q+1)(q^4+q^3+q^2+q+1) \\ &+ q(1+q)^4(3q^6+19q^5+36q^4+23q^3+36q^2+19q+3)\big] = \Phi_4(q)\cdot\sigma_2(V_{(3,2,1)}) \end{aligned}$$

is a polynomial identity (Lemma 11).

In particular, σ_2-G1 holds for both rank-2 shapes at $n = 6$:

$$\sigma_2(V_{(5,1)}) = q^3(1+q)^{18}(q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1),$$
$$\sigma_2(V_{(3,2,1)}) = q^9(1+q)^2\, P_{(3,2,1)}(q),$$

with $P_{(3,2,1)}(q) = q^{10} + 16q^9 + 105q^8 + 369q^7 + 759q^6 + 961q^5 + 759q^4 + 369q^3 + 105q^2 + 16q + 1$.

The proof of Theorem 3 occupies Sections 3 and 4. The two key polynomial identities (Lemmas 7 and 11) are finite checks that we verify both by hand and by SymPy (Section 7).

3 Shape $(5,1)$: the rank-1 leak between $V_{(5)}$ and $V_{(4,1)}$

3.1 The block decomposition

The branching $V_{(5,1)} \downarrow_{\mathfrak{S}_5} = V_{(5)} \oplus V_{(4,1)}$ gives a block-adapted basis $\{v_0;\, v_1, v_2, v_3, v_4\}$ of $V_{(5,1)}$, with $v_0 \in V_{(5)}$ and $v_1, \ldots, v_4 \in V_{(4,1)}$. The basis is chosen so that within $V_{(4,1)}$, the index v_1 is the standard tableau in which the box of 5 shares a column with the box of 6 (i.e. the single tableau where T_5 has its non-diagonal action), while v_2, v_3, v_4 are the three remaining $V_{(4,1)}$-tableaux. Set

$$U_{\text{swap}} := \operatorname{span}(v_0, v_1), \qquad U_{\text{stab}} := \operatorname{span}(v_2, v_3, v_4),$$

giving a $2+3$ split $V_{(5,1)} = U_{\text{swap}} \oplus U_{\text{stab}}$. Write the matrix of $M := \Pi_q^{(6)} \mid V_{(5,1)}$ in this basis as

$$M = \begin{pmatrix} A_{ss} & A_{st} \\ A_{ts} & A_{tt} \end{pmatrix},$$

where A_{ss} is 2×2, A_{st} is 2×3, A_{ts} is 3×2, and A_{tt} is 3×3.

Lemma 4 (Block ranks)**.** *In the basis above:*

(i) $\operatorname{rank}(A_{ss}) = 2$ (swap block has full rank)*;*

(ii) $\operatorname{rank}(A_{st}) = \operatorname{rank}(A_{ts}) = 1$ (rank-1 leak)*;*

(iii) $\operatorname{rank}(A_{tt}) = 1$*, hence* $\sigma_2(A_{tt}) = 0$*;*

(iv) $\operatorname{rank}(M) = 2 = \dim V_{(5,1)}^{H_6}$.

Proof. Direct computation in Hoefsmit's seminormal form. The block ranks (i)–(iv) are computed by row-reducing the explicit matrices in $\mathbb{Q}(q)$; `decomposition_test.py` outputs all four ranks. The full rank $\operatorname{rank}(M) = 2$ also matches the H-invariant theorem. □

3.2 The σ_2 split formula

Lemma 5 (σ_2 block split)**.** *For any block-decomposed matrix* $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$,

$$\sigma_2(M) = \sigma_2(A) + \sigma_2(D) + \operatorname{tr}(A)\operatorname{tr}(D) - \operatorname{tr}(BC). \tag{1}$$

Proof. $2\sigma_2(M) = (\operatorname{tr} M)^2 - \operatorname{tr}(M^2)$. Substitute $\operatorname{tr}(M) = \operatorname{tr}(A) + \operatorname{tr}(D)$ and $\operatorname{tr}(M^2) = \operatorname{tr}(A^2) + 2\operatorname{tr}(BC) + \operatorname{tr}(D^2)$. □

For shape $(5,1)$, applying (1) to $M = \Pi_q^{(6)}|V_{(5,1)}$ and using $\sigma_2(A_{tt}) = 0$ (Lemma 4(iii)):

$$\sigma_2(V_{(5,1)}) = \underbrace{\sigma_2(A_{ss})}_{=:[A]} + \underbrace{\operatorname{tr}(A_{ss})\operatorname{tr}(A_{tt}) - \operatorname{tr}(A_{st}A_{ts})}_{=:[B]}. \tag{2}$$

Proposition 6 (Closed forms for the swap and cross pieces)**.** *With the basis above, direct computation in Hoefsmit's seminormal form gives*

$$[A] = \frac{q^3(1+q)^{26}}{\Phi_4(q)},$$
$$[B] = \frac{q^4(1+q)^{18}\,(q^2+4q+1)\,(4q^4+9q^3+4q^2+9q+4)}{\Phi_4(q)}.$$

Computational verification. Each entry of the matrices $A_{ss}, A_{st}, A_{ts}, A_{tt}$ is an explicit element of $\mathbb{Q}(q)$ derived from the Hoefsmit form of $T_1, \ldots, T_5$. SymPy computes $\sigma_2(A_{ss})$ and the cross-trace $\operatorname{tr}(A_{ss})\operatorname{tr}(A_{tt}) - \operatorname{tr}(A_{st}A_{ts})$ symbolically; the resulting rational functions factor as displayed (`decomposition_test.py`). The denominator $\Phi_4(q) = q^2+1$ arises from the seminormal-form denominators $1/[5]_q = 1/(1+q+q^2+q^3+q^4)$ together with $1/[2]_q = 1/(1+q)$ collected from the various entries; the numerator factor Φ_4 in each piece is the irreducible cyclotomic that fails to cancel *piecewise* but cancels in the sum (see the next lemma). □

3.3 The cancellation identity

Lemma 7 ((5, 1) cancellation identity)**.** *The following polynomial identity holds in $\mathbb{Z}[q]$:*

$$(1+q)^8 + q(q^2+4q+1)(4q^4+9q^3+4q^2+9q+4) = (q^2+1)(q^6+12q^5+52q^4+88q^3+52q^2+12q+1). \quad (3)$$

Proof. Both sides are polynomials in q of degree 8. Expanding the left-hand side:

$$(1+q)^8 = 1+8q+28q^2+56q^3+70q^4+56q^5+28q^6+8q^7+q^8,$$
$$(q^2+4q+1)(4q^4+9q^3+4q^2+9q+4) = 4q^6+25q^5+44q^4+34q^3+44q^2+25q+4,$$
$$q\cdot(\text{above}) = 4q^7+25q^6+44q^5+34q^4+44q^3+25q^2+4q.$$

Summing,

$$\text{LHS} = q^8+12q^7+53q^6+100q^5+104q^4+100q^3+53q^2+12q+1.$$

On the right-hand side, expanding $(q^2+1)\cdot c(q)$ with $c(q) = q^6+12q^5+52q^4+88q^3+52q^2+12q+1$ gives

$$q^2\cdot c(q)+c(q) = q^8+12q^7+(52+1)q^6+(88+12)q^5+(52+52)q^4+(12+88)q^3+(1+52)q^2+12q+1.$$

Both sides agree term-by-term; LHS = RHS. □

Proof of Theorem 3(a). By Proposition 6 and (2),

$$\sigma_2(V_{(5,1)})\cdot\Phi_4(q) = q^3(1+q)^{26} + q^4(1+q)^{18}(q^2+4q+1)(4q^4+9q^3+4q^2+9q+4).$$

Factoring out $q^3(1+q)^{18}$ from the right,

$$\sigma_2(V_{(5,1)})\cdot\Phi_4(q) = q^3(1+q)^{18}\big[(1+q)^8 + q(q^2+4q+1)(4q^4+9q^3+4q^2+9q+4)\big].$$

By Lemma 7, $(1+q)^8 + q(q^2+4q+1)(4q^4+9q^3+4q^2+9q+4) = \Phi_4(q)\cdot c_{(5,1)}(q)$ with $c_{(5,1)}(q) = q^6+12q^5+52q^4+88q^3+52q^2+12q+1$, so

$$\sigma_2(V_{(5,1)}) = q^3(1+q)^{18}\,c_{(5,1)}(q).$$

The polynomial $c_{(5,1)}$ is palindromic with strictly positive integer coefficients $(1, 12, 52, 88, 52, 12, 1)$, $(1+q)^{18}$ is palindromic non-negative, and q^3 is a single positive monomial. Hence $\sigma_2(V_{(5,1)})$ is palindromic with non-negative integer coefficients, proving σ_2-G1 for $(5, 1)$ at $n = 6$. □

Remark 8 (Structural interpretation of (q^2+4q+1)). The palindromic factor (q^2+4q+1) in $[B]$ is the σ_1-*core of* $V_{(3,1)}$ *at* $n=4$, equivalently the σ_1-core of $V_{(3,1,1)}$ at $n=5$:

$$\mathrm{tr}(\Pi_q^{(4)} \mid V_{(3,1)}) = q(1+q)^2(q^2+4q+1), \quad \mathrm{tr}(\Pi_q^{(5)} \mid V_{(3,1,1)}) = q^3(1+q)^2(q^2+4q+1).$$

It also appears as the σ_2-core of both rank-2 shapes at $n=5$: $\sigma_2(\Pi_q^{(5)} \mid V_{(4,1)}) = q^3(1+q)^{12}(q^2+4q+1)$ and $\sigma_2(\Pi_q^{(5)} \mid V_{(3,2)}) = q^5(1+q)^8(q^2+4q+1)$. So this factor is a *universal* σ*-atom* that propagates by branching; the $[B]$ piece carries it via the rank-1 leak between $V_{(5)}$ and $V_{(4,1)}$. The $[A]$ piece, in contrast, is "purely new at $n=6$": the swap block A_{ss} couples $V_{(5)}$ with the T_5-active row of $V_{(4,1)}$ and contributes a clean $q^3(1+q)^{26}/\Phi_4$ that has no $V_{(5,1)}$-internal σ_1 analogue at lower n.

4 Shape $(3,2,1)$: branching with three summands

4.1 The branching block decomposition

The branching $V_{(3,2,1)} \downarrow_{\mathfrak{S}_5} = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ splits the basis of $V_{(3,2,1)}$ according to the position of the box of 6:

- 5 tableaux with 6 at $(2,0)$ form the $V_{(3,2)}$-block,
- 6 tableaux with 6 at $(1,1)$ form the $V_{(3,1,1)}$-block,
- 5 tableaux with 6 at $(0,2)$ form the $V_{(2,2,1)}$-block.

We use the $5+11$ split with $r = V_{(3,2)}$-block (rows/columns 0–4) and $e = V_{(3,1,1)} \oplus V_{(2,2,1)}$-block (rows/columns 5–15):

$$M = \Pi_q^{(6)} \mid V_{(3,2,1)} = \begin{pmatrix} A_{rr} & A_{re} \\ A_{er} & A_{ee} \end{pmatrix}, \qquad A_{rr} \in M_{5\times 5},\ A_{ee} \in M_{11\times 11}.$$

Lemma 9 (Block ranks for $(3,2,1)$)**.** *(i)* $\mathrm{rank}(A_{rr}) = 2$ (the $V_{(3,2)}$-block alone carries the full image rank)*;*

(ii) $\mathrm{rank}(A_{ee}) = 2$*;*

(iii) $\mathrm{rank}(A_{re}) = \mathrm{rank}(A_{er}) = 2$*;*

(iv) $\mathrm{rank}(M) = 2 = \dim V_{(3,2,1)}^{H_6}$.

The blocks individually have rank 2*, but the whole matrix* M *also has rank* 2*: the column spaces of all four blocks lie in a single* 2*-dimensional subspace, so the off-diagonal coupling forces the column contributions to be linearly dependent across the block partition.*

Proof. Direct computation by row-reduction in Hoefsmit's seminormal form (`decomp_321_v2.py`). Item (i) is the striking observation that the heaviest branching summand $V_{(3,2)}$ already carries the full rank-2 image of $\Pi_q^{(6)}$ — every column of M is a $\mathbb{Q}(q)$-linear combination of two columns supported on the $V_{(3,2)}$-rows. This is consistent with the dominance order: $(3,2)$ is the dominant summand of $V_{(3,2,1)} \downarrow$. □

4.2 The three-piece σ_2 decomposition

Applying Lemma 5 to M with the (rr, ee) split,

$$\sigma_2(V_{(3,2,1)}) = [A_{rr}] + [A_{ee}] + [\text{cross}], \tag{4}$$

where

- $[A_{rr}] \coloneqq \sigma_2(A_{rr})$ (the $V_{(3,2)}$-block second symmetric function),
- $[A_{ee}] \coloneqq \sigma_2(A_{ee})$ (the rest-block second symmetric function), and
- $[\text{cross}] \coloneqq \operatorname{tr}(A_{rr})\operatorname{tr}(A_{ee}) - \operatorname{tr}(A_{re}A_{er})$ (the cross-coupling).

By Lemma 9, all three quantities are nontrivial rational functions in q.

Proposition 10 (Closed forms for $(3,2,1)$ pieces)**.** *With the $5+11$ split above, direct computation in Hoefsmit's seminormal form gives the three explicit rational summands*

$$\begin{aligned}
[A_{rr}] &= \frac{q^9(1+q)^{10}\,(q^2+q+1)(q^2+4q+1)}{\Phi_4(q)},\\
[A_{ee}] &= \frac{q^{11}(1+q)^2\,(q^2+q+1)(q^2+4q+1)(q^4+q^3+q^2+q+1)}{\Phi_4(q)},\\
[\text{cross}] &= \frac{q^{10}(1+q)^6\,(3q^6+19q^5+36q^4+23q^3+36q^2+19q+3)}{\Phi_4(q)},
\end{aligned}$$

each having palindromic numerator with non-negative integer coefficients.

Computational verification. SymPy computation, `decomp_321_v2.py`. Sum verified to equal the full $\sigma_2(V_{(3,2,1)})$. □

4.3 The cancellation identity

Lemma 11 ($(3,2,1)$ cancellation identity)**.** *Let $p_1(q) = q^2+q+1$, $p_2(q) = q^2+4q+1$, $p_4(q) = q^4+q^3+q^2+q+1$, $p_6(q) = 3q^6+19q^5+36q^4+23q^3+36q^2+19q+3$, and let $P_{(3,2,1)}(q) = q^{10}+16q^9+105q^8+369q^7+759q^6+961q^5+759q^4+369q^3+105q^2+16q+1$. Then*

$$(1+q)^8\,p_1\,p_2 + q^2\,p_1\,p_2\,p_4 + q(1+q)^4\,p_6 = \Phi_4(q)\cdot P_{(3,2,1)}(q). \tag{5}$$

Proof. A polynomial identity in $\mathbb{Z}[q]$: both sides are polynomials of degree 12. The identity is verified directly via expansion (SymPy script `verify_proof_identities.py`, function `verify_identity_321`). The verification reduces to checking that the LHS expanded coefficient-wise matches the RHS: both sides are $q^{12}+16q^{11}+105q^{10}+369q^9+759q^8+961q^7+$(palindrome continuing)— the full coefficient list is the convolution of $\Phi_4 = (1,0,1)$ with $P_{(3,2,1)}$'s coefficient list $(1,16,105,369,759,961,759,369,105,16,1)$, giving a degree-12 palindrome with positive integer coefficients on both sides. This finite check passes. □

Proof of Theorem 3(b). By Proposition 10 and (4),

$$\sigma_2(V_{(3,2,1)})\cdot\Phi_4(q) = q^9(1+q)^{10}p_1p_2 + q^{11}(1+q)^2p_1p_2p_4 + q^{10}(1+q)^6p_6.$$

Factoring out $q^9(1+q)^2$ from the right-hand side gives

$$\sigma_2(V_{(3,2,1)})\cdot\Phi_4(q) = q^9(1+q)^2\big[(1+q)^8p_1p_2 + q^2p_1p_2p_4 + q(1+q)^4p_6\big].$$

By Lemma 11, the bracket equals $\Phi_4(q) \cdot P_{(3,2,1)}(q)$, so

$$\sigma_2(V_{(3,2,1)}) = q^9(1+q)^2\, P_{(3,2,1)}(q).$$

The polynomial $P_{(3,2,1)}$ is palindromic with positive integer coefficients, $(1+q)^2$ is palindromic non-negative, and q^9 is a single positive monomial. Hence $\sigma_2(V_{(3,2,1)})$ is palindromic with non-negative integer coefficients, proving σ_2-G1 for $(3,2,1)$ at $n=6$. □

Remark 12 (Structural interpretation of the $(3,2,1)$ pieces). The factor (q^2+4q+1) appears in two of the three pieces (in $[A_{rr}]$ and $[A_{ee}]$). It is the same universal σ_1-atom of $V_{(3,1,1)}$ at $n=5$ (Remark 8). The factor $(q^2+q+1) = \Phi_3(q)$ is the third cyclotomic, appearing as the σ_1-content of $V_{(2,2,1)}$: $\mathrm{tr}(\Pi_q^{(5)} \mid V_{(2,2,1)}) = q^4(1+q)^2$ does *not* carry it directly, but it appears via the $V_{(3,1,1)} \downarrow_{\mathfrak{S}_4} = V_{(3,1)} \oplus V_{(2,1,1)}$ deeper-branching mechanism. The factor $(q^4+q^3+q^2+q+1) = \Phi_5(q)$ arises from the $\mathfrak{S}_5$-axial-distance scaling $1/[5]_q$. The cross piece $p_6 = 3q^6+19q^5+36q^4+23q^3+36q^2+19q+3$ is a new palindromic atom not directly inherited from any single rank-1 shape; its palindromic non-negativity is a key structural fact.

5 The trivial rank-0 case $\lambda = (2,1^4)$

For completeness:

Proposition 13 (Trivial case)**.** $\Pi_q^{(6)} \mid V_{(2,1^4)} = 0$, *hence* $\sigma_2(V_{(2,1^4)}) = 0$.

Proof. $\ell((2,1^4)) = 5 > (n+1)/2 = 7/2$, so by the rank-killing criterion ($\ell(\lambda) > (n+1)/2 \Rightarrow V_\lambda^{H_n} = 0$), $\dim V_{(2,1^4)}^{H_6} = 0$. By the H-invariant theorem, $\mathrm{rank}(\Pi_q^{(6)} \mid V_{(2,1^4)}) = 0$, so the operator is identically zero. $\sigma_2 = 0$ is trivially palindromic non-negative. Direct Hoefsmit-form verification confirms $\Pi_q^{(6)} \mid V_{(2,1^4)} = 0$ (`verify_proof_identities.py`). □

6 Combined statement

Theorem 14 (σ_2-G1 at $n=6$, structurally)**.** *For every* $\lambda \vdash 6$ *with* $\dim V_\lambda^{H_6} \le 2$, $\sigma_2(\Pi_q^{(6)} \mid V_\lambda) \in q^a(1+q)^b \cdot \mathbb{Z}_{\ge 0}[q]$ *with palindromic core. Explicitly:*

(1) $\lambda \in \{(6),(4,1,1),(3,3),(2,2,2)\}$*: rank* ≤ 1*, so* $\sigma_2 = 0$.

(2) $\lambda \in \{(3,1^3),(2,2,1^2),(2,1^4),(1^6)\}$*: rank* 0*, so the operator is zero and* $\sigma_2 = 0$.

(3) $\lambda = (5,1)$*:* $\sigma_2 = q^3(1+q)^{18} c_{(5,1)}$ *with* $c_{(5,1)} \in \mathbb{Z}_{>0}[q]$ *palindromic (Theorem 3(a)).*

(4) $\lambda = (3,2,1)$*:* $\sigma_2 = q^9(1+q)^2 P_{(3,2,1)}$ *with* $P_{(3,2,1)} \in \mathbb{Z}_{>0}[q]$ *palindromic (Theorem 3(b)).*

The remaining shape $\lambda = (4,2)$ has rank 3 and is outside the scope of σ_2-G1 in the rank-2 sense; it requires separate treatment.

7 Computational scripts and verification

The following scripts under `~/projects/scratch/2026-04-25-hu-mathas-test/` reproduce all computations:

- `compute_filtration.py`: builds Hoefsmit seminormal-form matrices of $T_1,\dots,T_5$ on $V_{(5,1)}$ and verifies the Hecke quadratic, braid, and commutation relations symbolically.

- `decomposition_test.py`: computes the $2+3$ block decomposition of $\Pi_q^{(6)} \mid V_{(5,1)}$ in the swap/stable basis and produces the closed forms for $[A]$ and $[B]$.
- `decomp_321_v2.py`: same for $V_{(3,2,1)}$ in the $V_{(3,2)}$/rest basis, producing $[A_{rr}], [A_{ee}]$, [cross].
- `verify_proof_identities.py`: the master verification script checking Lemmas 7 and 11 as polynomial identities, and confirming Proposition 13.

All computations are over $\mathbb{Q}(q)$ with exact symbolic arithmetic.

8 Outlook: rank-3 and beyond

The proof of σ_2-G1 here uses a *rank-1-leak* structural decomposition: σ_2 is exhibited as a sum of pieces whose numerators are products of palindromic σ_1-atoms inherited from smaller-rank shapes via the $\mathfrak{S}_5$-branching, plus a single cyclotomic Φ_4 denominator that cancels via a finite polynomial identity. The pattern is:

- For $\lambda = (5,1)$: rank-1 leak between $V_{(5)}$ and $V_{(4,1)}$ contributes (q^2+4q+1) via the $V_{(4,1)}$-summand's σ_1-core;
- For $\lambda = (3,2,1)$: branching with three summands, contributions of (q^2+q+1), (q^2+4q+1), $(q^4+q^3+q^2+q+1)$ from different summand-pair couplings.

The next test is σ_3-G1 for the rank-3 shape $\lambda = (4,2)$ at $n=6$, where $\dim V_{(4,2)}^{H_6} = 3$ and the image is a 3-dimensional block. The cancellation will involve Φ_4 and likely Φ_5 denominators; the expected structure is a *rank-2-leak* generalization.

The structural meaning of the Φ_4 cancellation — whether it follows from a parity-purity/Sandvik-monodromic argument, from Goertzen–Williamson invariance of $(1+s)$-cones, or from a Soergel-bimodule decomposition of $\Pi_q^{(6)} = \mathrm{BS}(w_0)$ in EW Hecke category — is the most natural next research direction. The present proof is finite and explicit, but the cancellation identities cry out for an a-priori derivation.

References

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